

Cost effectiveness of ward based non-invasive ventilation for acute exacerbations of chronic obstructive pulmonary disease: economic analysis of randomised controlled trial.

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Evidence abstract

OBJECTIVE To evaluate the cost effectiveness of standard treatment with and without the addition of ward based non-invasive ventilation in patients admitted to hospital with an acute exacerbation of chronic obstructive pulmonary disease.

DESIGN Incremental cost effectiveness analysis of a randomised controlled trial.

SETTING Medical wards in 14 hospitals in the United Kingdom.

PARTICIPANTS The trial comprised 236 patients admitted to hospital with an acute exacerbation of chronic obstructive pulmonary disease and mild to moderate acidosis (pH 7.25-7.35) secondary to respiratory failure.

The economic analysis compared the costs of treatment that these patients received after randomisation.

MAIN OUTCOME MEASURE Incremental cost per in-hospital death.

RESULTS 24/118 died in the group receiving standard treatment and 12/118 in the group receiving non-invasive ventilation ($P=0.05$).

Allocation to the group receiving non-invasive ventilation was associated with a reduction in costs of 49362 pounds sterling (78741 dollars; 73109 euros), mainly through reduced use of intensive care units.

The incremental cost effectiveness ratio was -645 pounds sterling per death avoided (95% confidence interval -2310 pounds sterling to 386 pounds sterling), indicating a dominant (more effective and less costly) strategy.

Modelling of these data indicates that a typical UK hospital providing a non-invasive ventilation service will avoid six deaths and three to nine admissions to intensive care units per year, with an associated cost reduction of 12000-53000 pounds sterling per year.

CONCLUSIONS Non-invasive ventilation is a highly cost effective treatment that both reduced total costs and improved mortality in hospital.

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